Lab activity(DRL)

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Roll no:030

Task no 1(CompanyDB):

Queries

a) Delete employees hired before January 1, 2021

```sql
DELETE FROM Employees
WHERE HireDate < '2021-01-01';
```

b) List all departments where employees earn more than 60000

```sql
SELECT DISTINCT D.DepartmentName
FROM Departments D
JOIN Employees E ON D.DepartmentID = E.DepartmentID
WHERE E.Salary > 60000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Con

DepartmentName
Finance
HR

c) Departments where employees have salary > 65000 OR hired after Jan 1, 2022

```sql
SELECT DISTINCT D.DepartmentName FROM Departments D
JOIN Employees E ON D.DepartmentID = E.DepartmentID
WHERE E.Salary > 65000
UNION
SELECT DISTINCT D.DepartmentName
FROM Departments D
JOIN Employees E ON D.DepartmentID = E.DepartmentID
WHERE E.HireDate > '2022-01-01';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

DepartmentName
HR
Finance

d) Employees with salary > 60000

```sql
SELECT * FROM Employees
WHERE Salary > 60000;
```

Result Grid | Filter Rows: | Edit: | Export/In

EmployeeID	FirstName	LastName	DepartmentID	HireDate	Salary
4	Zara	Ali	3	2022-11-05	65000.00
5	Omar	Javed	1	2023-02-20	70000.00
NULL	NULL	NULL	NULL	NULL	NULL

e) Departments located in Karachi

```sql
SELECT * FROM Departments
WHERE Location = 'Karachi';
```

DepartmentID	DepartmentName	Location
1	HR	Karachi

f) Employees hired in 2020

```sql
SELECT FirstName, LastName
FROM Employees
WHERE YEAR(HireDate) = 2020;
```

FirstName	LastName
Ali	Khan

g) Average salary of all employees

```sql
SELECT AVG(Salary) AS AverageSalary
FROM Employees;
```

AverageSalary
60000.000000

h) Employees whose last name starts with 'A'

```sql
SELECT * FROM Employees
WHERE LastName LIKE 'A%';
```

EmployeeID	FirstName	LastName	DepartmentID	HireDate	Salary
2	Sara	Ahmed	2	2019-03-22	60000.00
4	Zara	Ali	3	2022-11-05	65000.00

i) Distinct department locations

```sql
SELECT DISTINCT Location
FROM Departments;
```

Result Grid | Filter Rows:

Location
Karachi
Lahore
Islamabad

j) Total number of employees in each department

```sql
SELECT D.DepartmentName, COUNT(E.EmployeeID) AS TotalEmployees
FROM Departments D
LEFT JOIN Employees E ON D.DepartmentID = E.DepartmentID
GROUP BY D.DepartmentName;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

DepartmentName	TotalEmployees
HR	2
IT	2
Finance	1

k) Employees hired after 2021

```sql
SELECT * FROM Employees
WHERE HireDate > '2021-12-31';
```

Result Grid | Filter Rows: | Edit: | Export/Imp

EmployeeID	FirstName	LastName	DepartmentID	HireDate	Salary
4	Zara	Ali	3	2022-11-05	65000.00
5	Omar	Javed	1	2023-02-20	70000.00

l) Maximum salary in the company

```sql
SELECT MAX(Salary) AS MaxSalary
FROM Employees;
```

Result Grid | Filter Rows: | E:

MaxSalary
70000.00

m) Minimum salary in each department

```sql
SELECT D.DepartmentName, MIN(E.Salary) AS MinSalary
FROM Departments D
JOIN Employees E ON D.DepartmentID = E.DepartmentID
GROUP BY D.DepartmentName;
```

DepartmentName	MinSalary
HR	50000.00
IT	55000.00
Finance	65000.00

Task no 2(Restaurant Menu Management)

Queries

a) Find all available menu items.

```sql
SELECT * FROM MenuItems
WHERE Available = TRUE;
```

ItemID	ItemName	Category	Price	Avai
1	Burger	Main Course	12.00	1
2	Pizza	Main Course	15.00	1
3	Caesar Salad	Appetizers	8.00	1
5	Coffee	Beverages	4.00	1

b) List all orders along with the total amount for each customer.

```sql
SELECT O.OrderID, C.CustomerName, O.TotalAmount
FROM Orders O
JOIN Customers C ON O.CustomerID = C.CustomerID;
```

OrderID	CustomerName	TotalAmount
1	Alice Williams	30.00
2	Bob Johnson	25.00
3	Charlie Brown	20.00
4	Diana White	18.00
5	Ethan Green	14.00

c) Find the average price of items in each category.

```sql
SELECT Category, AVG(Price) AS AveragePrice
FROM MenuItems
GROUP BY Category;
```

Category	AveragePrice
Main Course	13.500000
Appetizers	8.000000
Desserts	6.000000
Beverages	4.000000

d) Find the most expensive item on the menu.

```sql
SELECT * FROM MenuItems
WHERE Price = (SELECT MAX(Price) FROM MenuItems);
```

ItemID	ItemName	Category	Price	Available
2	Pizza	Main Course	15.00	1

e) List all orders placed by a specific customer (e.g., CustomerID = 2)

```sql
SELECT * FROM Orders
WHERE CustomerID = 2;
```

OrderID	CustomerID	OrderDate	TotalAmount
2	2	2024-07-11	25.00

f) Retrieve the total amount spent by each customer.

```sql
SELECT C.CustomerName, SUM(O.TotalAmount) AS TotalSpent
FROM Customers C
JOIN Orders O ON C.CustomerID = O.CustomerID
GROUP BY C.CustomerName;
```

CustomerName	TotalSpent
Alice Williams	30.00
Bob Johnson	25.00
Charlie Brown	20.00
Diana White	18.00
Ethan Green	14.00

g) Find the total quantity of each menu item sold.

```sql
SELECT M.ItemName, SUM(OI.Quantity) AS TotalQuantitySold
FROM MenuItems M
JOIN OrderItems OI ON M.ItemID = OI.ItemID
GROUP BY M.ItemName;
```

h) List all menu items that are available but have not been ordered.

```sql
SELECT M.ItemName
FROM MenuItems M
WHERE M.Available = TRUE
AND M.ItemID NOT IN (SELECT DISTINCT ItemID FROM OrderItems);
```

i) Calculate the total revenue from orders in July 2024.

```sql
SELECT SUM(TotalAmount) AS JulyRevenue
FROM Orders
WHERE MONTH(OrderDate) = 7 AND YEAR(OrderDate) = 2024;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content

JulyRevenue
107.00

Task 3(E-commerce Sales Analysis)

Queries:

a) Total sales amount for each product

```sql
SELECT P.ProductName, SUM(SI.Quantity * SI.ItemPrice) AS TotalSales
FROM SaleItems SI
JOIN Products P ON SI.ProductID = P.ProductID
GROUP BY P.ProductName;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ProductName	TotalSales
Smartphone	699.00
Laptop	1299.00
Coffee Maker	89.00
Book	15.00
Headphones	398.00

b) Top 3 products by total sales amount

```sql
SELECT P.ProductName, SUM(SI.Quantity * SI.ItemPrice) AS TotalSales
FROM SaleItems SI
JOIN Products P ON SI.ProductID = P.ProductID
GROUP BY P.ProductName
ORDER BY TotalSales DESC
LIMIT 3;
```

ProductName	TotalSales
Laptop	1299.00
Smartphone	699.00
Headphones	398.00

c) Customers who purchased more than $500

```sql
SELECT C.CustomerName, SUM(S.TotalAmount) AS TotalSpent FROM Customers C
JOIN Sales S ON C.CustomerID = S.CustomerID
GROUP BY C.CustomerName
HAVING SUM(S.TotalAmount) > 500;
```

CustomerName	TotalSpent
Alice Williams	798.00
Bob Johnson	1299.00

d) Products currently in stock

```sql
SELECT * FROM Products
WHERE Stock > 0;
```

ProductID	ProductName	Category	Price	Stock
1	Smartphone	Electronics	699.00	50
2	Laptop	Electronics	1299.00	20
3	Coffee Maker	Appliances	89.00	30
4	Book	Books	15.00	100
5	Headphones	Electronics	199.00	40

e) Average price of products in 'Electronics'

```sql
SELECT AVG(Price) AS AvgElectronicsPrice
FROM Products
WHERE Category = 'Electronics';
```

AvgElectronicsPrice
732.333333

f) Products in the 'Books' category

```sql
SELECT * FROM Products
WHERE Category = 'Books';
```

Result Grid | Filter Rows: | Edit:

ProductID	ProductName	Category	Price	Stock
4	Book	Books	15.00	100

g) Product with the lowest price

```sql
SELECT * FROM Products
WHERE Price = (SELECT MIN(Price) FROM Products);
```

Result Grid | Filter Rows: | Edit:

ProductID	ProductName	Category	Price	Stock
4	Book	Books	15.00	100

h) Customer details by email

```sql
SELECT * FROM Customers
WHERE Email ='jane.smith@example.com';
```

i) All sales made on '2024-07-01'

```sql
SELECT * FROM Sales
WHERE SaleDate = '2024-07-01';
```

Result Grid | Filter Rows: | Edit:

SaleID	CustomerID	SaleDate	TotalAmount
1	1	2024-07-01	798.00